Research Gap

#	Research gap	Evidence from the survey
1	**Computation speed and detection strength are studied separately.** Papers either optimise how fast a CRC is computed or analyse how well it detects errors, but rarely evaluate both together.	Speed and hardware only: rows 3, 5, 11. Detection analysis without implementation cost: rows 1, 6, 7, 14.
2	**Idealised error models.** Most detection analysis assumes random errors on an AWGN channel, whereas real links also suffer burst and correlated errors.	AWGN or random-error simulation: rows 2, 7, 10, 14. Only row 15 models fading and interference, and it addresses 5G-NR V2X only.
3	**Theoretical results are not validated under realistic networking conditions.** Few studies test CRC with realistic packet sizes, protocols or traffic.	Theory only: rows 1, 6. Simulation only: rows 2, 7, 10, 13, 14, 15.
4	**Lack of side-by-side comparison of CRC variants and simpler schemes.** CRC-8, CRC-16 and CRC-32 are rarely compared with each other, or with parity and checksum, under identical conditions.	Row 8 compares CRC with parity and sum check but is a review without experiments. Row 1 covers 5G NR polynomials only. Rows 5 and 11 cover speed only.
5	**The trade-off between overhead and reliability for short packets is not quantified for general data communication.**	Row 7 shows a rate loss from CRC at short blocklengths, and row 10 uses a single short configuration (8-bit CRC, 16 information bits). Neither studies CRC length against packet size.
6	**CRC miss rate is not linked to retransmission performance.** The effect of undetected errors on ARQ throughput, delay and retransmission count is seldom analysed.	Row 9 surveys ARQ and HARQ without CRC-specific analysis. Rows 4 and 13 treat replay as a backstop, and row 12 uses CRC only as a trigger.
7	**Adaptive CRC selection is under-explored.** Matching the CRC to channel conditions appears in only one setting, and its benefit fades at higher mobility and SNR.	Row 15 only, in simulation.

8	**Limited reproducibility.** Few simulation-based studies report open, reusable tools for comparing CRC performance.	Among the error-detection studies in the survey, row 10 is the one built on open-source code.

Problem statement

Existing studies of CRC either optimise its computation speed or analyse its detection capability under idealised channel assumptions, and few evaluate detection performance, overhead and processing cost together under both random and burst errors. Direct comparison of standard CRC variants with simpler schemes such as parity and checksum is also limited, as is the link between undetected errors and retransmission performance. This project analyses CRC-8, CRC-16 and CRC-32 against parity and checksum across random and burst-error channels to quantify these trade-offs.

Objectives

1. Implement CRC-8, CRC-16 and CRC-32 together with parity and checksum error detection.
2. Simulate transmission over a random-error channel (binary symmetric) and a burst-error channel (Gilbert-Elliott).
3. Measure detection rate, undetected-error probability, redundancy overhead and computation time for each scheme.
4. Evaluate the effect of undetected errors on ARQ throughput and the number of retransmissions.
5. Recommend which scheme suits which packet size and error condition.